

# AC4100

## Management Accounting: Planning and Control

### Winter 2022 Examination

Paper 1 · 1.5 hours · Answer Question 1 (compulsory) and EITHER Question 2 OR Question 3

|                    |                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module             | AC4100 — Management Accounting: Planning and Control                                                                                                     |
| Session            | Winter 2022                                                                                                                                              |
| Structure          | Question 1 is compulsory (40 marks). Then answer EITHER Question 2 OR Question 3 (60 marks each). Total 100 marks.                                       |
| This scheme covers | Question 1 in full, PLUS both Question 2 and Question 3 in full — so you can study whichever optional route you'd pick, or compare both before deciding. |

**Own-figure rule.** If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the method marks for those later steps. This applies throughout the multi-product breakeven workings in Question 1.

**Question 1**

Compulsory — Multi-Product Breakeven, Regression Analysis &amp; Definitions (40 Marks)

**Part (i)****15 Marks**

Smith Company can produce two types of carpet cleaners, Brighter and Cleaner. Data on these two products are as follows:

|                       | Brighter | Cleaner |
|-----------------------|----------|---------|
| Sales volume in units | 400      | 600     |
| Unit sales price      | €960     | €880    |
| Unit variable cost    | 200      | 700     |

The number of machine hours to produce one unit of Brighter is 1, while the number of machine hours for each unit of Cleaner is 2. Total fixed costs for the manufacture of both products are €290,000.

**Required:**

1. Determine the breakeven point in total units for Smith Company, based on the assumption that the sales mix (on the basis of relative sales volume in units) remains constant. Use the weighted-average contribution margin approach.
2. At this breakeven level, how many units of each product must be sold?
3. Briefly identify any conclusion that can be drawn from your analysis.

**FULL MODEL ANSWER**

**1. Breakeven point in total units (8 marks)**

|                                     | Brighter   | Cleaner    |
|-------------------------------------|------------|------------|
| Unit sales price                    | 960        | 880        |
| Unit variable cost                  | (200)      | (700)      |
| <b>Contribution margin per unit</b> | <b>760</b> | <b>180</b> |

Sales mix (relative unit volume) = 400:600 = 40% Brighter : 60% Cleaner

Weighted-average CM per unit = (40% × €760) + (60% × €180)

= €304 + €108

**= €412 per unit**

Breakeven (total units) = Total fixed costs ÷ Weighted-average CM per unit

= €290,000 ÷ €412

**≈ 703.88 units (round up to 704 units to guarantee breakeven is reached)**

**2. Units of each product at breakeven (4 marks)**

|          | Calculation  | Units    |
|----------|--------------|----------|
| Brighter | 703.88 × 40% | ≈ 281.55 |
| Cleaner  | 703.88 × 60% | ≈ 422.33 |

Cross-check: contribution at these breakeven volumes = (281.55 × €760) + (422.33 × €180) = €290,000 exactly, confirming the split is correct.

**3. Conclusion (3 marks)**

**The breakeven mix is achievable within existing capacity, but only barely illustrates a deeper limitation of the method.** At the breakeven volumes above, machine hours required are  $(281.55 \times 1) + (422.33 \times 2) \approx 1,126$  hours, comfortably below the 1,600 hours implied by the company's actual current sales volumes (400 and 600 units) — so capacity isn't the binding constraint at breakeven itself. The more important conclusion is that this whole approach rests entirely on the assumption that the 40:60 sales mix stays fixed regardless of volume, which is rarely true in practice: if the actual mix shifts toward the lower-margin Cleaner as volume changes, the true breakeven point would be higher than 704 units, and if it shifts toward the higher-margin Brighter, the true breakeven would be lower. The weighted-average approach is only as reliable as the constant-mix assumption underpinning it.

**How the 15 Marks Are Likely Allocated**

| Tier         | Marks | What separates this tier                                                                                                                                                                                                                  |
|--------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(1) 8</b> | 8     | Correctly calculates each product's contribution margin per unit, correctly derives the 40:60 sales mix weights from unit volumes, and correctly calculates the weighted-average CM and resulting breakeven of $\approx 704$ total units. |
| <b>(2) 4</b> | 4     | Correctly splits the breakeven total using the 40:60 mix to reach $\approx 282$ Brighter and $\approx 422$ Cleaner units, with own-figure credit if the total breakeven carried forward is wrong but the split method is right.           |
| <b>(3) 3</b> | 3     | Identifies a genuine limitation or insight from the analysis (e.g. the constant sales mix assumption, or the capacity/machine-hours implication) rather than simply restating the numbers already calculated.                             |

Dr. Ross Kelly owns a GP practice in North Dublin. They have noticed a significant change in their cost behaviour over the past 3 years, partially due the increased use of digitalization as a result of the Covid-19 Pandemic. Dr. Kelly's accountant, Ms. Mary Carrol, has used regression analysis to help understand the drivers of the practice's costs and has had the following results:

Total Costs for patient =  $a + b_1x_1 + b_2x_2 + b_3x_3$   
 Where  $a$  = constant  
 $x_1$  = the length of time of the consultation (mins)  
 $x_2$  = In person consultation (1 – in person; 0 – telephone)  
 $x_3$  = Prescription needed (=1 if true, 0 if false)  
 $a$ ,  $b_1$ ,  $b_2$  and  $b_3$  are coefficients of the regression model.

The research is based on 2,360 consultations over the past 12 months and showed the following regression results:

*R-squared: 62%*

*Constant term: €40*

|             | Length of consultation | In person | Prescription |
|-------------|------------------------|-----------|--------------|
| Coefficient | €0.5                   | €8        | €3           |
| t-value     | 10.80                  | 4.73      | 2.43         |

### Part (ii)(1)

7 Marks

1. What is the estimated cost for a patient whose consultation took 12 minutes in an in-person appointment, and required a prescription?

## FULL MODEL ANSWER

$$\text{Total cost} = a + b_1x_1 + b_2x_2 + b_3x_3$$

$$= 40 + (0.5 \times 12) + (8 \times 1) + (3 \times 1)$$

$$= 40 + 6 + 8 + 3$$

$$= \text{€57}$$

Both  $x_2$  and  $x_3$  take the value 1 here, since the consultation was in person (not telephone) AND required a prescription — both conditions in the question apply simultaneously, so both coefficients are added in full, not averaged or chosen between.

### How the 7 Marks Are Likely Allocated

| Tier | Marks | What separates this tier                                                                                            |
|------|-------|---------------------------------------------------------------------------------------------------------------------|
| 6-7  | 6-7   | Correctly substitutes all four values into the regression equation and reaches €57.                                 |
| 3-5  | 3-5   | Correct equation set up but one term omitted or a dummy variable incorrectly set to 0 instead of 1 (or vice versa). |
| 0-2  | 0-2   | No attempt, or the equation fundamentally misapplied.                                                               |

#### Part (ii)(2)

**3 Marks**

2. Identify the dummy variables in the equation.

### FULL MODEL ANSWER

The dummy variables are  **$x_2$  (in person consultation)** and  **$x_3$  (prescription needed)** — both are binary, taking only the value 1 or 0 depending on whether the condition is true.

$x_1$  (length of consultation in minutes) is **not** a dummy variable — it's a continuous variable that can take any numerical value, not just 0 or 1.

### How the 3 Marks Are Likely Allocated

| Tier | Marks | What separates this tier                                                                                                       |
|------|-------|--------------------------------------------------------------------------------------------------------------------------------|
| 3    | 3     | Correctly identifies both $x_2$ and $x_3$ as the dummy variables, and (ideally) notes $x_1$ is continuous rather than a dummy. |
| 1-2  | 1-2   | Identifies one of the two dummy variables correctly.                                                                           |
| 0    | 0     | No attempt, or $x_1$ incorrectly identified as a dummy variable.                                                               |

**Part (ii)(3)****5 Marks**

3. Comment on how reliable this model is in predicting costs.

**FULL MODEL ANSWER****R-squared of 62% suggests moderate, not strong, explanatory power.**

The model explains 62% of the variation in per-patient costs, leaving 38% unexplained by the three variables included. That's a reasonable fit for a cost model with only three explanatory variables, but it's far from a complete picture — other genuine cost drivers (specific treatment or test type, patient complexity or comorbidities, staff mix on a given day, other practice overheads not captured by consultation-level detail) are likely missing from the model.

**The t-values indicate all three variables are statistically significant, but by differing margins.**

With a very large sample (2,360 consultations), even a fairly low t-value can be statistically significant. Length of consultation ( $t=10.80$ ) and in-person status ( $t=4.73$ ) are both very strongly significant, comfortably clearing conventional thresholds. Prescription needed ( $t=2.43$ ) is the weakest of the three — still significant at the 5% level, but noticeably less strongly than the other two, meaning there's more uncertainty around exactly how much a prescription adds to cost.

**Statistical significance is not the same as full reliability for cost prediction.**

A variable being statistically significant confirms the relationship almost certainly isn't due to chance, but it says nothing about whether the model is precise or complete enough to rely on for detailed cost prediction or pricing decisions — the 38% of unexplained variation matters for that purpose regardless of how significant the included variables are.

**The model reflects a specific practice over a specific, unusual period.**

Because the data covers the past 12 months at one practice during a period of “significant change in cost behaviour” driven partly by pandemic-related digitalisation, the relationships estimated may not be stable going forward, or may not transfer to a different practice with a different patient mix or cost structure — a real limitation on how far the model's predictions should be trusted outside the specific context it was estimated from.

**How the 5 Marks Are Likely Allocated**

| <b>Tier</b> | <b>Marks</b> | <b>What separates this tier</b>                                                                                                                                                                                                                                                                                  |
|-------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4-5</b>  | 4-5          | Correctly interprets the 62% R-squared as moderate (not weak, not strong) explanatory power, correctly assesses the relative strength of the three t-values, and distinguishes statistical significance from full predictive reliability, ideally also flagging the model's specific, possibly unstable context. |
| <b>2-3</b>  | 2-3          | Correctly interprets R-squared or the t-values but not both, or doesn't distinguish significance from overall reliability.                                                                                                                                                                                       |
| <b>0-1</b>  | 0-1          | No attempt, or a fundamental misinterpretation of R-squared or t-values (e.g. treating 62% as a poor fit, or treating any positive t-value as proof of full model reliability).                                                                                                                                  |



**Part (iii)****10 Marks**

Define the following terms:

- Learning Curve Effect.
- Economic Order Quantity.
- Decreasing returns to scale.

**FULL MODEL ANSWER**

**1. Learning Curve Effect.** The observed phenomenon whereby the time (and therefore cost) required to complete a repetitive task falls by a consistent percentage each time cumulative output doubles, as workers become more practised and efficient at the task through repetition. It's most relevant for new, labour-intensive, or complex processes early in a product's life, and tends to flatten out once workers reach a stable, experienced level of efficiency (the steady-state phase).

**2. Economic Order Quantity (EOQ).** The order quantity that minimises the total cost of managing inventory, by balancing the two opposing cost pressures of ordering (which favours large, infrequent orders to spread the fixed cost of placing each order) and holding (which favours small, frequent orders to minimise the amount of stock sitting in the warehouse at any time). EOQ is calculated as the square root of  $(2 \times \text{annual demand} \times \text{cost per order} \div \text{holding cost per unit})$ , and represents the order size at which total ordering costs and total holding costs are exactly equal.

**3. Decreasing returns to scale.** A situation in production where, as all inputs are increased by a given proportion, output increases by a smaller proportion — i.e. doubling every input less than doubles output. This typically arises once a firm grows beyond its efficient scale, as coordination, communication, and management complexity begin to outweigh the benefits of further expansion, in contrast to increasing returns to scale (where output grows faster than inputs) or constant returns to scale (where output grows in exact proportion to inputs).

**How the 10 Marks Are Likely Allocated**

| <b>Tier</b> | <b>Marks</b> | <b>What separates this tier</b>                                                                                         |
|-------------|--------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>8-10</b> | 8-10         | All three terms defined precisely and completely, with the mechanism behind each explained (not just a one-line label). |
| <b>5-7</b>  | 5-7          | Two terms defined precisely; the third vague, partial, or missing.                                                      |
| <b>2-4</b>  | 2-4          | One term defined precisely, or all three defined only in vague, imprecise terms.                                        |
| <b>0-1</b>  | 0-1          | No attempt, or definitions that don't distinguish these terms from unrelated concepts.                                  |

**Question 1 Total: 40 Marks**

**Question 2****Absorption Costing, Cost Allocation & Activity-Based Costing (60 Marks)**

In a general manufacturing environment, the following routine has been followed for several years to arrive at an estimate of the price for a contract. The process of estimating is commenced by referring to a job cost card for some previous similar job and evaluating the actual materials and direct labour hours used on that job at current prices and rates. Production overheads are calculated and applied as a percentage of direct wages. The percentage is derived from figures appearing in the accounts of the previous year, using the total production overhead cost divided by the total direct wages costs. One-third is added to the total production overhead cost to cover administrative charges and profit.

**Part (a)****20 Marks**

**(a)** Discuss the merits and deficiencies of a conventional absorption costing approach when used to determine the full cost of a product, which may be subsequently used as part of a cost plus formula for establishing selling price.

**FULL MODEL ANSWER**

---

## Merits

**Simplicity and ease of application.** A single overhead absorption rate (here, a percentage of direct wages) is straightforward to calculate and apply, requiring no elaborate cost-driver analysis — well suited to a business estimating prices for one-off contracts quickly, as described in the scenario.

**It ensures full cost recovery in principle.** By building fixed production overheads into the unit cost, and adding a further margin for administration and profit, the resulting price is designed to cover all the costs of the business plus a margin, in contrast to a marginal-cost-only approach that risks under-pricing if fixed costs aren't separately recovered.

**It complies with external financial reporting requirements.** Absorption costing (unlike marginal costing) is the basis required for inventory valuation under financial reporting standards, so a business already using it for costing purposes has one consistent cost base for both pricing and external reporting.

## Deficiencies

**A single, wages-based overhead rate can badly distort product costs.** Absorbing all production overhead as a percentage of direct wages assumes overhead consumption is proportional to labour cost — but overheads are often driven by very different factors (machine hours, number of setups, batch sizes, order complexity). A labour-light, machine-intensive job would be systematically under-costed under this approach, while a labour-intensive job would be over-costed, regardless of their actual overhead consumption.

**Basing the rate on last year's figures assumes stability that may not hold.** Using the previous year's actual overhead-to-wages ratio to set this year's rate assumes cost structures and activity levels haven't materially changed — a poor assumption in a period of changing costs, inflation, or shifting production volumes, and one that bakes last year's inefficiencies or one-off costs into every price quoted this year.

**The one-third mark-up for admin and profit is arbitrary and inflexible.** A flat, fixed mark-up applied regardless of the specific contract, market conditions, or competitive pressure takes no account of what the market will actually bear for a given job — the business may price itself out of contracts it could have profitably won at a lower margin, or leave money on the table on contracts where it could have charged more.

**Full cost data can create a false sense of precision for pricing decisions.** A single “full cost” figure derived this way looks precise but is built on a chain of simplifying assumptions (a single overhead base, last year's rate, a flat mark-up) — treating it as an accurate reflection of what a specific contract actually costs to fulfil risks poor pricing decisions dressed up in apparently rigorous numbers.

A strong answer connects the deficiencies specifically back to the scenario described (a single wages-based absorption rate, last year's figures, a flat one-third mark-up) rather than listing generic absorption costing criticisms in the abstract.



**How the 20 Marks Are Likely Allocated**

| <b>Tier</b>  | <b>Marks</b> | <b>What separates this tier</b>                                                                                                                                                                                                                                    |
|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>15-20</b> | 15-20        | Covers multiple genuine merits AND deficiencies, explicitly connecting the deficiencies to the specific routine described in the scenario (single wages-based rate, prior-year figures, flat mark-up), rather than a generic absorption-vs-marginal costing essay. |
| <b>10-14</b> | 10-14        | Covers several merits and deficiencies accurately but in generic terms, without tying them specifically to the scenario given.                                                                                                                                     |
| <b>5-9</b>   | 5-9          | One-sided discussion (mostly merits, or mostly deficiencies), or a superficial treatment of both.                                                                                                                                                                  |
| <b>0-4</b>   | 0-4          | No attempt, or a fundamental misunderstanding of absorption costing.                                                                                                                                                                                               |

**Part (b)****20 Marks**

**(b)** Outline the important features of any successful scheme of allocating, apportioning and absorbing indirect costs to products.

**FULL MODEL ANSWER**

**Costs should be traced as directly as possible before any apportionment is needed.** Wherever an indirect cost can genuinely be identified with a specific cost centre or department without estimation (allocation, in the strict sense), it should be assigned that way rather than lumped into a general pool and apportioned using an arbitrary basis — the more costs that can be directly allocated rather than apportioned, the less distortion the scheme introduces.

**The apportionment basis chosen should genuinely reflect how the cost is actually incurred.** Where a cost genuinely can't be traced directly and must be shared across cost centres (apportionment), the basis used (floor area for rent, headcount for canteen costs, machine value for insurance) should have a real causal or logical connection to why the cost arises, rather than being chosen simply because the data happens to be readily available.

**The absorption base used to bring overheads into product costs should reflect what actually drives overhead consumption.** A successful scheme uses an absorption base (machine hours, labour hours, or another driver) that genuinely correlates with how much overhead a product actually causes to be incurred — a single labour-hour or wages-based rate applied uniformly across genuinely different production processes (as in the scenario for part (a)) is a common source of distorted product costs precisely because it fails this test.

**Different cost centres or departments may need different absorption rates.** A single, plant-wide overhead rate assumes every part of the business consumes overhead in the same way — a successful scheme recognises that a highly automated department (where machine hours drive overhead) needs a different absorption base from a labour-intensive department (where labour hours drive overhead), rather than forcing one rate to fit the whole business.

**The scheme should be reviewed and updated regularly, not fixed indefinitely.** Cost structures, activity volumes, and production methods change over time — a scheme based on figures from several years ago (as in the scenario) risks embedding outdated assumptions into every current costing decision. Regularly revisiting both the overhead pools and the bases used keeps the scheme relevant to current operating conditions.

**The scheme should be transparent and consistently applied.** Whatever the chosen approach, applying it consistently across products and periods (rather than varying the method opportunistically) is essential for the resulting costs to be comparable and genuinely useful for decision-making, rather than being a black box that management can't sense-check or challenge.

**The cost and complexity of the scheme itself should be proportionate to the benefit it delivers.** A more granular, activity-based approach (see part (c)) generally produces more accurate product costs, but also costs more to design, implement and maintain — a successful scheme matches its level of sophistication to genuine business need, rather than pursuing precision for its own sake regardless of the cost of achieving it.

A strong answer moves through the full chain (allocation → apportionment → absorption) explicitly, rather than treating "cost allocation" as a single undifferentiated step, since the question specifically names all three stages.



**How the 20 Marks Are Likely Allocated**

| <b>Tier</b>  | <b>Marks</b> | <b>What separates this tier</b>                                                                                                                                                                                                                                                                                                         |
|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>15-20</b> | 15-20        | Addresses allocation, apportionment, AND absorption as distinct stages, identifies multiple genuine features of a good scheme for each (causal basis, departmental variation, regular review, proportionate cost/complexity), and connects the discussion to genuine costing principles rather than generic "good practice" statements. |
| <b>10-14</b> | 10-14        | Covers several genuine features but doesn't clearly distinguish allocation, apportionment and absorption as separate stages, or covers only one or two of the three.                                                                                                                                                                    |
| <b>5-9</b>   | 5-9          | General discussion of overhead costing without specific reference to what makes a scheme "successful" beyond restating the mechanics.                                                                                                                                                                                                   |
| <b>0-4</b>   | 0-4          | No attempt, or a fundamental misunderstanding of allocation, apportionment, or absorption.                                                                                                                                                                                                                                              |

**Part (c)****20 Marks**

**(c)** “An ABC system is justified whenever the costs of installing and operating the new system are more than offset by the long-term benefits that would be derived from the new ABC system.” (*ABC: The right approach for you?* - Robin Cooper)

Discuss briefly THREE Key benefits of using an ABC (Activity Based Costing) system.

**FULL MODEL ANSWER**

**1. More accurate product costs, particularly for businesses with diverse products or complex processes.** By tracing overheads to the specific activities that actually drive them (machine setups, order processing, quality inspections), rather than spreading overhead across products using a single volume-based measure like labour or machine hours, ABC avoids the systematic cross-subsidisation that a conventional system produces — high-volume, simple products are typically over-costed under conventional absorption costing (as illustrated in part (a)), while low-volume, complex products are under-costed, because the simple products get charged for a disproportionate share of overhead they didn't actually cause. ABC corrects this by linking cost to actual activity consumption.

**2. Better information for pricing and product mix decisions.** Because ABC reveals which products or services are genuinely profitable once overhead is fairly attributed (rather than which look profitable only because conventional costing under-costed them), management gets a much sounder basis for pricing decisions and for deciding which products, customers, or contracts to prioritise, drop, or promote — directly addressing the cost-plus pricing distortion risk identified in part (a).

**3. Improved visibility of cost drivers supports genuine cost control and process improvement.** ABC forces the organisation to identify and quantify the actual activities that consume resources (number of setups, purchase orders processed, inspections performed), which highlights specifically where non-value-adding activity or inefficiency exists in a way a single overhead percentage never could. This visibility supports targeted process improvement and cost reduction efforts aimed at the actual drivers of cost, rather than a blanket cost-cutting exercise with no clear sense of where the waste actually sits.

The Robin Cooper quote frames ABC adoption as a cost-benefit decision, not a universal prescription — a complete answer could briefly acknowledge that ABC's benefits (above) have to be weighed against genuinely higher implementation and ongoing data-collection costs, meaning ABC is most clearly justified for businesses with diverse products and complex, non-volume-driven overheads, and least justified for simple, homogeneous operations where a conventional system may already give a reasonably accurate picture at much lower cost.

**How the 20 Marks Are Likely Allocated**

| <b>Tier</b>  | <b>Marks</b> | <b>What separates this tier</b>                                                                                                                                                                                                                                          |
|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>15-20</b> | 15-20        | Discusses three genuinely distinct benefits (cost accuracy, pricing/mix decisions, cost driver visibility for process improvement) each with a clear explanation of the underlying mechanism, ideally connecting back to the cost distortion problem raised in part (a). |
| <b>10-14</b> | 10-14        | Three benefits named but with thinner explanation, or significant overlap between the three (e.g. two benefits that are really the same point phrased differently).                                                                                                      |
| <b>5-9</b>   | 5-9          | One or two genuine benefits discussed adequately.                                                                                                                                                                                                                        |
| <b>0-4</b>   | 0-4          | No attempt, or a fundamental misunderstanding of what ABC actually does differently from conventional absorption costing.                                                                                                                                                |

**Question 2 Total: 60 Marks**

**Question 3****Rethink Limited — Budgeting Objectives & Devolved Budgetary Control (60 Marks)**

You have recently been appointed as Management Accountant for Rethink Limited a registered charity that provides advice and financial assistance to pensioners. Rethink employs 68 staff across 14 locations throughout Ireland and has annual spending of €7,200,000. Rethink has no formal budgeting systems other than an annual central review of spend at each of its locations.

Rethink has applied to a government department for €100,000 in funding. You have been asked to prepare for a forthcoming meeting with this government department, at which Rethink's objectives will be evaluated. Rethink plans to spend this sum installing a system of devolved budgetary control whereby responsibility for cost control and budget delivery will be devolved to operational staff throughout the organisation's fourteen locations.

**Part (a)****30 Marks**

**(a)** Explain THREE objectives of budgeting.

**FULL MODEL ANSWER**

---

**1. Planning — forcing the organisation to think systematically about the future.** The budgeting process requires Rethink's management to translate its strategic objectives (advising and financially assisting pensioners) into specific, quantified plans for the coming period — how much will be spent, where, and on what. For an organisation that currently has “no formal budgeting systems other than an annual central review,” simply going through this planning exercise properly for the first time would itself be a significant step forward, forcing each of the fourteen locations to articulate what it actually needs and why, rather than spend being reviewed only after the fact.

**2. Control — providing a benchmark against which actual performance and spending can be assessed.** A budget sets a clear, agreed-in-advance standard for what spending at each location and activity should be, against which actual results can be compared throughout the year, not just once annually. Variances between budgeted and actual spend flag where corrective action is needed promptly, rather than being discovered only in an annual central review after the money has already been spent — this is particularly valuable for a charity accountable to funders and donors for how its money is actually used.

**3. Coordination — ensuring the activities of different parts of the organisation fit together consistently.** With 68 staff spread across 14 separate locations, there's a real risk that each location plans and spends in isolation, potentially duplicating effort, competing for the same limited central resources, or pursuing inconsistent priorities. A properly integrated budgeting process brings each location's plans together into a single, coordinated organisational plan, so that Rethink's finite €7.2 million annual spend (plus the requested €100,000) is allocated coherently across the whole charity rather than as fourteen independent, uncoordinated decisions.

Other valid objectives include **communication** (conveying organisational priorities and expectations down to operational staff at each location) and **motivation** (particularly relevant here, since devolving budget responsibility to location staff, as Rethink proposes, is intended specifically to motivate more engaged and accountable local cost management) — either would also be creditable as one of the three objectives, and motivation in particular connects naturally into part (b) below.

**How the 30 Marks Are Likely Allocated**

| <b>Tier</b>  | <b>Marks</b> | <b>What separates this tier</b>                                                                                                                                                                                                                                                                                                                      |
|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>23-30</b> | 23-30        | Explains three genuinely distinct objectives of budgeting in depth (from planning, control, coordination, communication, motivation), with each explanation specifically connected to Rethink's actual situation (multiple locations, no formal budgeting system currently, charity funder accountability) rather than generic textbook definitions. |
| <b>15-22</b> | 15-22        | Three objectives explained accurately but in generic terms, without connecting them to Rethink's specific circumstances.                                                                                                                                                                                                                             |
| <b>8-14</b>  | 8-14         | Two objectives explained adequately, or three named with minimal explanation.                                                                                                                                                                                                                                                                        |
| <b>0-7</b>   | 0-7          | No attempt, or fewer than two genuine objectives identified.                                                                                                                                                                                                                                                                                         |

**Part (b)****30 Marks**

**(b)** Explain THREE human/behavioural factors to be considered in the design of the budgetary control system to help ensure a successful devolution of responsibility to location budget holders.

**FULL MODEL ANSWER**

**1. Genuine participation of location staff in setting their own budgets, not just being handed a target.** Devolving budget *delivery* responsibility to operational staff at each of Rethink's fourteen locations is unlikely to succeed if the budgets themselves are still set centrally and simply imposed on local staff — genuine participation in setting the targets they'll be held accountable for builds commitment and buy-in, and also draws on local staff's first-hand knowledge of what each location actually needs, which central office may not have. Purely imposed, top-down budgets risk being seen as unrealistic or unfair by the people now expected to deliver them, undermining the whole point of devolving responsibility in the first place.

**2. Setting targets that are challenging but genuinely achievable, to avoid demotivation or dysfunctional behaviour.** Budget targets set unrealistically tight (perhaps to please the funding government department) risk demotivating location staff who see the target as unattainable from the outset, while targets set too loosely provide no real cost control benefit at all. Poorly calibrated targets can also encourage dysfunctional behaviour — for instance, a location manager under pressure to hit an unrealistic budget might defer necessary spending on pensioner services near year-end purely to hit the number, which runs directly against Rethink's actual charitable mission.

**3. Clear, fair, and non-punitive use of budget variances, so accountability doesn't create fear.** How management responds to variances between budgeted and actual spend at each location matters enormously for whether devolved responsibility is embraced or resented. If variances are used purely to assign blame rather than to understand genuine causes (which may include factors outside a location manager's control, such as unexpected local demand for services), budget holders will become defensive, may start manipulating figures to avoid unfavourable variances, or may become reluctant to flag emerging problems early — the opposite of the transparency and accountability the devolved system is meant to achieve.

Other valid factors include **adequate training and support** for location staff who may have no prior budgeting experience (relevant here since Rethink has “no formal budgeting systems” currently, meaning location staff are unlikely to have practised this skill before) and **linking the reward and recognition system** to budget performance in a way that reinforces the new devolved responsibility rather than undermining it — either would also be creditable as one of the three factors.

**How the 30 Marks Are Likely Allocated**

| <b>Tier</b>  | <b>Marks</b> | <b>What separates this tier</b>                                                                                                                                                                                                                                                                                                                        |
|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>23-30</b> | 23-30        | Explains three genuinely distinct human/behavioural factors (participation, target-setting difficulty, fair use of variances, training, reward alignment) in depth, with each explanation specifically connected to Rethink's situation (no prior budgeting experience, fourteen dispersed locations, charitable mission, government funder scrutiny). |
| <b>15-22</b> | 15-22        | Three factors explained accurately but in generic terms, without connecting them to Rethink's specific circumstances.                                                                                                                                                                                                                                  |
| <b>8-14</b>  | 8-14         | Two factors explained adequately, or three named with minimal explanation.                                                                                                                                                                                                                                                                             |
| <b>0-7</b>   | 0-7          | No attempt, or fewer than two genuine behavioural factors identified.                                                                                                                                                                                                                                                                                  |

**Question 3 Total: 60 Marks**